

Exploratory Behavior of Two Species of Murid Rodents, *Acomys cahirinus* and *Mus musculus*: A Comparative Study

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The exploratory behavior of two species of murid rodents, *Acomys cahirinus* and *Mus musculus*, was compared in four experiments: In the first, the responses of the two species to a novel arena were studied. *Mus* was found to take longer to enter the arena, and to spend more time in the relatively familiar or safer start box, than was *Acomys*. The results suggest that *Acomys* may persevere longer in exploring particular areas, whereas *Mus* appear to explore in the open arena by using frequent shifts of attention. The second experiment investigated species differences in response to the addition of a small novel object. Although the species did respond differently, the major species differences seemed to be related more to the open arena than to the object. The third experiment tested the hypothesis that both species would explore more if there was somewhere to hide (e.g., an artificial burrow) than if there was not. It was found that *Acomys* treated the available artificial burrow as another novel object, while *Mus*, as predicted, spent more time hiding inside it than did *Acomys*. The fourth experiment investigated burrow use when a model "predator" was introduced: Both species increased their use of the burrow but some species differences were found. *Mus* responded to the model more by freezing, or running immediately into the burrow; *Acomys* responded more by fleeing. © 1985 Academic Press, Inc.

Until recently, much of our knowledge of animal exploratory behavior was based upon laboratory studies of the rat. In recent years, this knowledge

has become slowly broadened in two important directions: First, studies of foraging behavior have contributed information concerning the ways in which animals utilize space and resources in familiar habitats: and, second, more comparative studies of animal exploration have emerged, particularly in relation to captive animals and their needs (see Birke & Archer, 1983, for review).

In the study reported here, two species of murid rodents were compared in a series of experiments designed to investigate responses to unfamiliar environments. One of these two species was the Spiny Mouse, *Acomys cahirinus*, which is a small murid rodent inhabiting arid regions of the Middle East (Dieterlen, 1962). *Acomys* is increasingly being used in laboratory studies of brain and behavior, being used particularly in studies of early behavioral development. It is unique among the Muridae in that it produces precocial offspring, and has thus been used, for example, in studies of olfactory imprinting (e.g., Porter and Etscorn, 1974). It has also been used in comparative studies of brain morphology. For example, D'Udine and Gozzo (1983) reported that *Acomys* had a substantially thicker hippocampal cortex than that of laboratory rodents of similar size with which it was compared, such as *Mus musculus*.

In view of this increasing interest, it is important to build up behavioral data for this species. Furthermore, although rather little is known about it, *Acomys* is, like a number of other small rodents, economically important in agricultural areas of its range. A primary aim of the present study was, therefore, to investigate some of the exploratory responses of *Acomys* in laboratory conditions. In the experiments reported here, the exploratory behavior of *Acomys* in unfamiliar surroundings was studied, and compared to those of another laboratory-bred species, *M. musculus*, about whose behavior rather more is known.

Given the rather different type of habitat to which these two species are adapted, differences between them might be expected in the responses shown to novel environments. Wild *Acomys*, for example, generally live on rocky outcrops or in arid sandy areas with little vegetation (Dieterlen, 1962), and so might be expected to be more ready to move across open spaces than *Mus*. In a study of the behavior of a number of murid rodent species, Wilson, Vacek, Lanier, & Dewsbury (1976) noted that rearing behavior was more common in desert-dwelling species, perhaps because it affords better all-round vision. On this basis, *Acomys* might be expected to show more rearing behavior than would *Mus*.

These predictions were tested in the series of experiments reported here. In the first two experiments, the responses of the two species to a novel environment were compared. It was predicted that *Acomys* would be more likely than *Mus* to leave the start box and to move into the center of the arena. In a third experiment, the two species were similarly exposed to a novel arena, but one containing an artificial "burrow" in

which animals could hide. The extent to which the two species would, in fact, use this burrow if danger threatened was tested in a fourth experiment, in which animals were exposed to a model "predator."

GENERAL METHODS

Sixteen animals of each species were used, of ages between 60 and 90 days. The colony of *Acomys* maintained at the Istituto di Psicobiologia e Psicofarmacologia, Rome originated from animals obtained from Parslow's Hamster Farm, Great Bookham, Surrey, England. The *Mus* were random-bred Swiss strain, obtained from Morini, Italy. Females were not included in this study since the estrous cycles of the two species are not comparable [e.g., the mean cycle length of *Acomys* females is 11.1 days (Peitz, 1981), whereas that of *Mus* is 4 days]. Thus the experiments were restricted to males in order to simplify the experimental designs.

Animals were housed in single-sex groups of eight *Mus*, or four to five *Acomys*, in cages $22 \times 12 \times 18$ cm. Cages were supplied with food and water ad libitum, and had sugar beet shavings as litter. Lighting in the stock room was reversed, with lights off from 0900 to 2100. During lights off, some illumination was supplied with dim red light. The room was at a constant temperature of 22°C, and a constant humidity of 60%.

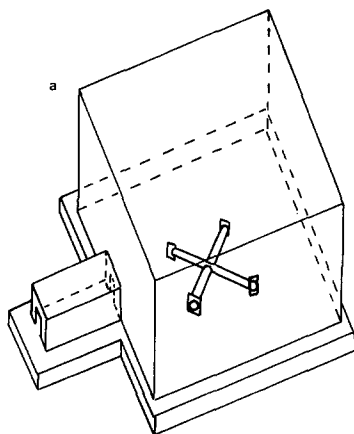
The testing apparatus was housed in the same room, and consisted of a square arena, $54 \times 54 \times 30$ cm, made of black Perspex, with clear Perspex lid and front wall, through which the animals could be viewed. The floor was made of clear Perspex, covering a paper surface on which squares were marked out. To the left of the arena (see Fig. 1) a start box could be attached. This was $20 \times 10 \times 7$ cm, with a sliding door at one end to allow entry to the arena. This also had a clear Perspex top, so that both start box and arena were exposed to similar illumination from above.

Between tests of different animals, the floor (plus any objects used in specific experiments) was washed down with 70% alcohol, and air dried, to remove the majority of odor traces.

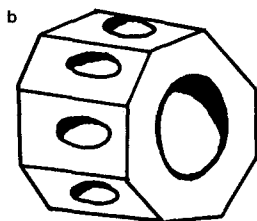
Prior to testing, the animal to be tested was placed in the start box, and allowed 1 min to familiarize itself with the start box before the test began. The door to the arena was then raised (and left raised in all experiments except Experiment 4), and recording commenced. All recording was carried out by two observers, each using a separate keyboard, both of which operated a 20-channel Esterline Angus event recorder. The categories used are described below for each experiment. Testing was carried out under the dim red background light of the stock room, as described above. There was also background white noise from the room ventilator.

STATISTICS

Parametric statistics were used throughout. However, since the measures of latency and duration were not always normally distributed, the data



SCALE 1:100



SCALE 1:50



SCALE 1:20

FIG. 1. (a) Testing arena ($54 \times 54 \times 30$ cm) and start box ($20 \times 10 \times 7$ cm). In Experiment 3 and 4 a cross made of steel tubing (diameter 4 cm) with four arms of equal length (10 cm) was placed in the center. (b) The object consisted of a hexagonal piece of white plastic, with a hole through the middle and small holes on each face. Each hole was approximately 1.5 cm in diameter. (c) A hawk-shaped piece of black Perspex, approx 20 cm in length.

were normalized by log transforming them prior to statistical analysis. For frequencies, the statistics were carried out on the raw data. Tests used included *t* tests (paired and unpaired) and two-way analyses of variance in conjunction with post hoc Duncan tests, as appropriate.

EXPERIMENT 1

In this experiment, the responses of the two species to an unfamiliar open arena were assessed. Animals could enter the arena from a start box on one side, which remained open throughout the experiment, thus providing a relatively "safe" place in which to hide. The arena was not completely open, however: In order to prevent animals from moving straight across the middle of the arena, a central square made of clear Perspex was incorporated, which could only be entered by one of four small holes. Thus, in order to explore the central area, the animal had to take the possible risk of leaving the walls and entering the central square.

Methods

The central square was made of clear Perspex, 30 × 30 cm, placed centrally in the arena. Each side had a circular hole, 2.5 cm in diameter, with the center 2.75 cm above the base of the square. The sides of the square were parallel to the sides of the arena. Each animal was placed as described above into the start box, and, once recording had begun, its behavior was observed for a 10-min period, as follows:

1. *Frequencies.* The frequencies of the following behavior patterns were scored, to provide a measure of the readiness of the animal to leave the box or to enter the central square:

—peeping, in which the animal "peeps" out of the start box. This was recorded when the head actually emerged from the start box, but the animal did not emerge completely.

—stretching. This is analogous to the "stretched attention" posture described for rats by Van Der Poel (1979), in which the hind legs remain stationary, while the snout and front legs are stretched forward in an apparently "tentative" manner.

—head poke, in which the animal pokes its head into one of the holes in the central square. Head pokes were only scored, however, when they were *not* followed by entering the square.

In addition, the frequencies of rearing, jumping, and grooming behavior were recorded.

2. *Duration/Latencies.* The latencies taken to leave the start box, and to enter the central square, were also recorded. Also, the durations of time spent in the start box and the central square were recorded as well as the duration of time spent grooming any part of the body.

As soon as each animal was tested, it was returned to its home cage, and the arena was cleaned as previously described.

Results

The results of this experiment are shown in Tables 1 and 2. Table 1 gives the mean values ($\pm$ SE) of all measures obtained in the experiment, and their associated probabilities. Table 2 provides measures of the behavior of the animals during the initial 2 min: that is, for the square, the total time spent in the square in the first 2 min after having entered it. These were included in order to provide some indication of how animals distributed their time in either the square or the box.

Data from this experiment were analyzed using *t* tests. On frequency measures, *Mus* were significantly more likely than *Acomys* to show peeping ($p = .001$) and stretching ($p < .01$), but less likely to show jumping ($p = .02$). *Mus* were also more likely to show head poking into the central square ($p = .05$). On duration measures, the statistical tests were performed on the log-transformed data, as noted above. For these, the latency to leave the box was not significant.

The duration of time spent in the square showed no significant differences, while the duration in the start box showed that *Mus* spent significantly longer than did *Acomys* ($p = .05$). Grooming also showed significant differences: *Acomys* spent significantly longer time grooming than did *Mus* ($p < .05$).

Discussion

It is clear from these results that there are a number of species differences in responses to an unfamiliar arena. *Mus*, for example, spent longer in the start box throughout the test, and showed more peeping out, stretching,

TABLE 1
Summary of Results Obtained in "Open Field": Experiment 1

Behavioral measure	<i>Mus</i>	<i>Acomys</i>	<i>p</i>
1. Frequencies			
Rearing	73.2 $\pm$ 6.9	83.6 $\pm$ 6.8	NS
Peeping	5.8 $\pm$ 1.3	0.4 $\pm$ 0.08	0.001
Stretching	4.25 $\pm$ 0.4	0	<0.01
Head poke into square	11.68 $\pm$ 1.3	7.4 $\pm$ 1.4	0.05
Jumping	0.06 $\pm$ 0.001	3.56 $\pm$ 1.4	0.02
Grooming	4.8 $\pm$ 1.5	5.25 $\pm$ 0.4	NS
2. Duration(s)			
Latency to leave box	15.25 $\pm$ 1.6	9.4 $\pm$ 1.5	NS*
Latency to enter square	269.7 $\pm$ 55.0	495.0 $\pm$ 25.1	<0.001
Duration in box	132.2 $\pm$ 22.3	85.7 $\pm$ 11.1	0.05
Duration in square	40.1 $\pm$ 11.7	31.9 $\pm$ 7.0	NS
Groom	9.3 $\pm$ 3.2	18.0 $\pm$ 3.9	<0.05

Note. Results are expressed as means $\pm$ SE.

* Statistical tests on measure of durations performed on log-transformed data: see text.

TABLE 2
Summary of Behavior in Initial 2-min Period

	<i>Mus</i>	<i>Acomys</i>
Time spent in box as percentage of total time in box	23.7	23.1
Time spent in square as percentage of total time in square	39.5	86.5
Time in box (s) (mean)	31.3	19.8
Time in square (s) (mean)	15.8	27.6

and head poking into the central square. These are behavior patterns that suggest ambivalence: Peeping out, for instance, involves the animal in putting its head briefly outside the start box and then withdrawing it again. Stretching behavior is similarly ambivalent, as suggested for the rat by Van Der Poel (1979). Thus, in general, *Mus* is, as predicted, rather more "tentative" about leaving the relative familiarity or safety of the start box and venturing out into the open field than is *Acomys*.

On the basis of the longer time taken by *Mus* to leave the start box, it might be expected that they would, similarly, take longer to enter the square than would *Acomys*. This was not, in fact, the case: Rather, *Acomys* took significantly longer to enter the central square than did *Mus*. However, *Acomys* showed significantly fewer head pokes than did *Mus*. This suggests that, while *Acomys* take longer on average to enter the square, once they have made the decision they are less "ambivalent" about it. *Mus* typically preceded their initial entry into the square with several head pokes, which presumably provide them with some additional information about the square before taking the risk of entering it.

It is not clear from these data why *Acomys* should take so much longer than *Mus* to enter the square in the first place. Although we have no specific data on the location of the animals at different times throughout the test, *Acomys* appeared to move about the arena in a different manner from that of *Mus*. More specifically, *Acomys* gave the impression of having "somewhere to go": They seemed to move rapidly and purposefully around the outer square, often for several consecutive circuits. In these rapid laps of the circuit, these animals seemed rarely distracted by other stimuli, such as the holes into the central square. *Mus*, by contrast, gave the impression of being more tentative, and attending to a wider range of stimuli from the moment they first entered.

The data shown in Table 2 show considerable interspecific differences in the time spent in the square in the first 2 min after having entered it (expressed as percentage of the total time in the square). These data suggest that *Acomys* tend to explore the central square in a "burst" of exploratory activity immediately after the initial entry, whereas *Mus* tend to distribute their activity more evenly throughout the testing period. This observation, taken in conjunction with the impression noted above,

might suggest that the strategy of exploration adopted by *Acomys* might be one of exploring one area at a time, before moving onto another area, whereas that of *Mus* might be to sample briefly all available areas. This would be an interesting species difference, and could be further tested by experimentally providing the animals with a range of available areas to investigate.

Grooming behavior also showed significant differences between the two species. Although the frequencies with which this behavior was shown did not differ, the durations of grooming did. This difference could be an artifact of the method of testing, since no observations were made of grooming in the start box, which had opaque sides. However, it is unlikely that this explains the observed difference, given that no differences in grooming frequency were observed in this arena. This suggests that there is a difference in the mean duration of any bout of grooming, such that *Mus* tend to exhibit shorter bouts of grooming in the test situation than do *Acomys*.

In general, the results of this experiment supported our preliminary predictions concerning the behavior of these two species of rodent when confronted with a novel environment. However, this experiment does not clearly separate effects due to the novelty of separate features of the environment and effects due to the openness of the field. The observed responses may be due principally to the openness of the arena (exploration of which, in the wild, could lead to detection by predators) or they may simply represent a series of responses to novel stimuli per se. A second experiment was therefore conducted, designed to replicate the first experiment, but with the addition of a novel stimulus, a small, hexagonal object placed at the center of the square. The animals' responses both to the open arena and also to the novel object could then be recorded to determine the effects of adding a novel object on the animal's responses to the open arena.

EXPERIMENT 2: RESPONSES TO A NOVEL OBJECT

Methods

The apparatus and testing procedure were generally as described for Experiment 1, with the exceptions outlined below. The experiment utilized a two-by-two design, such that half the *Mus* and half the *Acomys* were tested with the object present ($N = 8$ for each species), and the other half of each species were tested without the object ($N = 8$ for each). The order of testing was randomized with respect to both species and presence or absence of object.

The object consisted of a hexagonal piece of white plastic, with a hole through the middle and small holes on each face, as shown in Fig. 1b. Each hole was approximately 1.5 cm in diameter. The object was placed

at the center of the square prior to testing. After each test, it was removed and placed in an alcohol solution, as described above.

The behavior patterns recorded were as in Experiment 1, except that (i) grooming was not recorded; and (ii) behavior directed towards the object was included. The categories of object-directed behavior were:

—sniff object: This was recorded whenever the vibrissae were within approximately 1 cm of the object's surface.

—peep object. This was recorded whenever the animal entered its snout into one of the holes in the object;

—bite object; and

—drag object. This was recorded whenever the animal bit the object and then held it in its mouth, dragging it along the floor of the arena.

In addition, the duration of sniffing the object was recorded.

Results

As in Experiment 1, statistical tests were performed on log-transformed data for measures of duration or latency, and on the raw data for measures of frequency. Data were initially analyzed using either a two-way analysis of variance or a *t* test, depending on the form of the data. The results of this experiment are summarized in Table 3.

TABLE 3
Summary of Results of Experiment 2: Presence or Absence of Novel Object

Species object	<i>Mus</i>		<i>Acomys</i>	
	With	Without	With	Without
Behavior				
Rear	68.9 ± 14.4	88.9 ± 13.2	75.1 ± 14.6	71.1 ± 8.9
Peep	5.4 ± 1.7	5.9 ± 2.1**	0.6 ± 0.5	0.12 ± 0.08
Stretch	2.7 ± 1.7	2.5 ± 0.7**	0	0
Head poke	10.1 ± 1.6	10.0 ± 1.4	14.5 ± 5.2	14.5 ± 3.4
Sniff object	8.9 ± 2.3	—	8.1 ± 1.9	—
Peep object	5.2 ± 2.4	—	2.9 ± 1.0	—
Bite object	0.12 ± 0.08	—	2.7 ± 1.5	—
Drag object	0	—	1.12 ± 0.6	—
Sniff duration	26.7 ± 9.3	—	32.5 ± 10.6	—
Latency to leave box	12.0 ± 3.9	12.7 ± 3.0**	3.5 ± 0.8	4.1 ± 1.8
Latency to enter square	201.7 ± 59.8	268.2 ± 80.6*	344.5 ± 70.7	442.0 ± 67.9
Duration in box	192.2 ± 58.7	131.1 ± 31.2	128.0 ± 42.4	125.0 ± 18.9
Duration in square	69.5 ± 21.3	52.2 ± 22.3	73.6 ± 15.0*	31.5 ± 12.4

Note. Results are expressed as means ± SE.

* *p* < .05.

** *p* < .01.

In general, similar effects were found in this experiment as were found in Experiment 1. Significant species differences were found on measures of peeping ($F(1, 31) = 14.72, p < .01$) and stretching ($F(1, 31) = 22.21, p < .01$), and on the latency to leave the start box ($F(1, 31) = 15.16, p < .01$), and to enter the central square ($F(1, 31) = 5.92, p < .05$). No significant differences emerged, however, when the presence/absence of the object was considered as a source of variation, on any measure.

Object-directed behaviors were analyzed using *t* tests. Although the mean values of some of these behaviors indicated differences, none of these were statistically significant.

Discussion

Overall, the results of this experiment repeat the findings of Experiment 1; as before, *Mus* show more tentative approaches to the arena, and are slower to leave the start box, whereas *Acomys* are slower to enter the central square. There appears, however, to be rather little effect of the object on measures of exploratory behavior: From the analyses of variance, it would seem that the principal source of significant variation is that due to species.

However, the presence of the object was not completely ignored by the animals. The data from measures of the time spent in the square, for example, suggest that this measure is affected by the presence of the object for *Acomys*, but not for *Mus*. *Acomys* seem to spend approximately twice as much time in the central square when the object is present than when it is not. A post hoc *t* test performed on the log-transformed data for *Acomys* only indicated that this difference was significant ($p < .05$), whereas a similar analysis on the data for *Mus* confirmed that there is no significant difference due to the object.

Although none of the behavior directed toward the object showed significant differences, the data do indicate trends that are noteworthy. First, peeping into the holes in the object was somewhat more common among *Mus* than among *Acomys*. Second, attempts to *do* something with the object (i.e., biting or dragging it) were almost entirely made by *Acomys*: Only one *Mus* tried to bite the object at all. This would suggest that there are some species differences in the ways that the object is dealt with, although the differences were not significant in this experiment. For *Acomys*, the problem would seem to be one of "What can I do with this object?", whereas for *Mus*, it is one of "What information can I gain from this object?" In other words, *Acomys* seem more concerned to manipulate the object directly than simply to investigate it superficially. Whether or not this is generally true across a range of different types of objects is, however, a matter for further investigation.

The results of this experiment, then, support the results of the first experiment, which demonstrated significant species differences on a number

of behavior measures. These differences are not, however, substantially affected by the introduction of a novel, rather conspicuous object, but appear primarily to represent responses to the open field. In the wild, small novel objects *may* be a source of danger [as they sometimes are for species of rodents commensal with humans, for whom novel objects may mean baits or traps (Cowan, 1983), so that moving out of a relatively safe place to explore an open field may carry greater risk (e.g., from predators)]. Thus, if confronted with an unfamiliar open area a small rodent will—if it is to avoid being eaten—need to know more about the general layout of the area than it will need to know about specific stimuli within it. Once it is fairly familiar with the general topography it is likely to be relatively safe from predators, not least because it will know where to hide (Glickman and Morrison, 1969; Metzgar, 1967).

However, this relative safety does depend upon the availability of possible places to hide: A familiar, but totally open space, with no hiding places, is just as dangerous as a less familiar one that offers some opportunity to hide or to escape. In the next experiment, the responses of the same two species were reexamined with a different kind of “novel object,” one which afforded an artificial “burrow.” It was predicted that both *Mus* and *Acomys* would tend to explore the arena more if they could return to the burrow as a “safe” place than if they could not. Since the benefits to be gained by exploration have to be weighed against the possible costs of predation, it might be less risky to explore an open arena in which there clearly are places to hide. Experiment 3 was designed to investigate this possibility further.

EXPERIMENT 3: USE OF ARTIFICIAL “BURROW”

Methods

A design similar to that employed in the previous experiment was used: That is, half the animals of each species were presented with the “burrow” which they could enter, and the other half were presented with the tube burrow but it was sealed off to prevent entry. This was done to control for the effect of the novelty of the tube *per se*. As before, the order of testing was randomized with respect to species and to whether or not the tube was open.

The tube burrow was made of steel, in a cross, with arms of equal length. The total length of the tube from end to end was approximately 25 cm, each end having a small lip for support. Thus, the lower surface of the tube was 1.5 cm from the arena floor. The inside of the tube was approximately 4 cm in diameter throughout, and all four arms joined in the middle so that an animal could leave the tube in a different direction from that in which it entered, or even turn around and leave from the same direction. The location of the tube in the arena is shown in Fig.

1a. No central square was used in this experiment and the ends of the tube were clearly visible from the start box.

When animals were tested in the closed-tube situation, the ends of the tube were sealed with adhesive tape. Matt, black tape was used for this, as it did not reflect much, and approximated the black appearance of the dark interior of the open tube.

The methods of testing were as previously described, and similar categories of behavior were scored: Categories of behavior directed toward the tube were head poking into the ends of the tube, sniffing the outer surface, biting any part of the tube, and entering the tube. Since the central square was not used in this experiment, latency and duration of square entry was not applicable in this experiment. Instead, the latency to contact the tube and the duration of time spent inside the tube (if the tube was open) were substituted.

Results

The results of this experiment are summarized in Table 4. Data were, as before, analyzed either by a two-way analysis of variance, or by a *t* test, depending on the form of the data. These analyses showed significant effects on frequency measures due to species for peeping ($F(1, 31) = 9.33$; $p < .01$), for stretching ($F(1, 31) = 8.23$; $p < .01$), for head poking

TABLE 4
Summary of Results of Experiment 3: Use of Tube "Burrow"

	<i>Mus</i>		<i>Acomys</i>	
	Open	Closed	Open	Closed
Behavior				
Rear	26.5 $\pm$ 5.3	36.2 $\pm$ 10.5	24.7 $\pm$ 6.7	25.2 $\pm$ 3.3
Peep	5.4 $\pm$ 2.5	3.9 $\pm$ 1.6**	0	0.1 $\pm$ 0.08
Stretch	1.12 $\pm$ 0.4	1.5 $\pm$ 0.8**	0	0
Head poke tube	3.75 $\pm$ 1.8	— *	12.0 $\pm$ 1.5	—
Sniff tube	38.5 $\pm$ 8.4	47.6 $\pm$ 6.3**	64.2 $\pm$ 5.4	60.2 $\pm$ 2.8
Bite tube	0	0	3.9 $\pm$ 1.1	9.2 $\pm$ 2.8
Enter tube	9.6 $\pm$ 2.4	—	9.4 $\pm$ 2.9	—
Sniff duration	125.4 $\pm$ 29.2	114.6 $\pm$ 6.4**	195.5 $\pm$ 13.0	209.4 $\pm$ 31.3
Entry duration	81.9 $\pm$ 27.8	—	29.7 $\pm$ 9.4	—
Latency to leave box	8.25 $\pm$ 2.4	33.0 $\pm$ 13.0**	5.6 $\pm$ 2.7	5.5 $\pm$ 1.2
Latency to contact tube	16.9 $\pm$ 6.7	47.5 $\pm$ 16.6**	7.1 $\pm$ 3.1	15.0 $\pm$ 4.9
Duration in box	196.2 $\pm$ 59.0	165.2 $\pm$ 45.9	99.6 $\pm$ 27.4	105.0 $\pm$ 16.3

Note. Results are expressed as means $\pm$ SE.

* $p < .05$.

** $p < .01$.

into the tube ($p < .01$, t test), and for sniffing the tube surface ($F(1, 31) = 6.57$, $p < .05$). There was a significant effect due to treatment (the state of the tube) on the frequency of biting the tube ($F(1, 31) = 19.4$; $p < .01$). On measures of latency or duration, significant effects due to species were found for duration of sniffing ($F(1, 31) = 8.6$, $p < .01$), latency to leave box ($F(1, 31) = 9.42$, $p < .01$), and latency to contact tube ($F(1, 31) = 8.75$, $p < .01$). The mean entry durations (shown in Table 4) were considerably different for the two species; *Mus* spent more than twice as long in the tube than did *Acomys* on average. However, the variance in this measure was considerable, and this difference did not reach significance ($t = 1.32$, for which p is approximately .1).

Discussion

The results of this experiment show a number of species differences, and also indicate that the two species do differ in their responses to the state of the tube (i.e., its availability as cover). First, *Acomys* showed more investigation of the outside of the tube than did *Mus*, whether it was open or closed. Second, *Mus* spent somewhat more time inside the tube than did *Acomys* (although, as pointed out above, this did not reach statistical significance). Third, *Acomys* showed more head poking without immediately entering the tube than did *Mus*. Finally, some differences emerged in relation to the closure of the tube. *Acomys*, for example, showed more biting if the tube was closed than if it was open. If the tube was closed, both species took considerably longer to make initial contact with it than if it was open, and *Mus* took very much longer to leave the starting box if the tube was closed.

These results lend support to the prediction that, if the open arena also contains a place in which it might be possible to take cover, *Mus* might be more exploratory than if it does not. *Mus* took very much longer to leave the start box and to make initial contact with the tube when it was closed than when it was open. This does not, of course, show conclusively that the significant variable was the openness of the ends, since the closed ends will inevitably appear different—for example, in color and contour—than the open ends. However, taken in conjunction with the observation that *Mus* spend more time inside the tube than do *Acomys*, it seems plausible that the animals—and *Mus* in particular—are more prepared to explore the arena when the object it contains offers a chance to hide.

In this experiment, *Acomys* were found to head poke into the tube significantly more often than *Mus*. Yet, in the previous experiments, the reverse was the case for head poking into the central square. This reversal may simply reflect a *Mus* preference for potential burrows. On the other hand, it may be that *Acomys* are simply more “tentative” in their exploration if they cannot see directly into an object. Head poking in the

previous experiments was through holes into a clear Perspex central square, into which, presumably, the animal could see anyway.

This experiment, then, suggests that exploratory responses may be affected by the kinds of resources available. In this experiment, a tube burrow was provided that, when open, provided potential cover. In the next experiment, a similar approach was used to determine the responses of the animals to possible predators. If the open tube had been perceived as offering a place to hide, then it would be predicted that, in the presence of a model predator, the animals would use the tube burrow more than they otherwise would.

In the present experiment, *Mus* used the burrow more than did *Acomys*, while *Acomys* explored the outside of the tube more than did *Mus*. On this basis, it might be predicted that, upon introduction of the "predator," the responses of the two species might differ. It might be expected, for instance, that *Acomys* would show more fleeing responses than would *Mus*, whereas the latter might be expected to use the burrow as a hiding place more than might *Acomys*.

EXPERIMENT 4: RESPONSES OF MUS AND ACOMYS TO A MODEL "PREDATOR"

Methods

Sixteen animals of each species were used, as before. The apparatus and metal tube burrow were used as they were for Experiment 3. However, in this experiment, the tube burrow was always used open. There were two other significant differences in methods from Experiment 3: First, in this experiment, the animals were in the apparatus for a total of 4 min; and second, once they had left the start box, the entrance to it was closed. Thus, the only available hiding place was the tube burrow.

For testing, each animal was placed in the start box and left for 1 min, as described above. After this, the door was raised and the animal was allowed to enter the arena, and then the door closed behind it. Observations were then made on the animal in the absence of the model for a period of 2 min. After 2 min, the predator model was introduced. This consisted of a "hawk"-shaped piece of black Perspex, approx 20 cm in length (depicted in Fig. 1c. It was attached to a piece of wire, and was moved slowly across the roof of the arena from side to side. It was not moved evenly, but was moved so as to resemble a hovering predator. This was achieved by slight vibration of the mounting wire. The behavioral responses of small mammals to overhead predators can depend on whatever behavior is ongoing at the time the predator is perceived (Fentress, 1968). Hence, the predator model was only introduced when the animal was actively moving outside the tube. This model was chosen on the basis of preliminary trials which indicated that, when it was in motion, it continued to elicit flight or immobility from both species throughout the 2-min period.

The predator model was moved continuously for the next 2 min, during which the behavior of the animals was recorded. The following categories of behavior were noted in each 2-min period. Freezing: The duration of any immobility, or "freezing," was recorded; Squares crossed: The number of squares of the open field crossed by the animal was recorded throughout. In addition, particular note was taken of the square crossed in the first 30 s after the model was presented, as a measure of whether or not the animal fled from the model.

The frequency of rearing and the frequency of tube entries were also scored, as well as the total duration of tube entry.

Results

The results of this experiment are summarized in Table 5. The results were analyzed using a two-way analysis of variance, with repeated measures. These analyses showed that, first, freezing duration showed a significant effect of the repeated measure (before vs after the model): ($F(1, 63) = 12.34, p < .01$). That is, both species froze significantly more often after the model. Second, the squares crossed showed a significant effect due to species ($F(1, 63) = 16.5, p < .01$). For this measure, the repeated measures factor did not show a significant effect, although the interaction was significant ($F(1, 63) = 12.3, p < .01$). A post-hoc Duncan test indicated that this was principally due to the treatment effect in *Acomys*, which, unlike *Mus*, showed a considerable increase in the number of squares crossed during the predator presentation ($p < .01$). This difference is also indicated by the difference in mean number of squares crossed in the 30-s period immediately after the model was presented (*Mus* = 6.1, *Acomys* = 19.4). The frequency of tube entries showed

TABLE 5
Summary of Results from Experiment 4: Response to Predator Model

	<i>Mus</i>		<i>Acomys</i>	
	Before	After	Before	After
Behavior				
Freeze duration(s)				
(total)	0.125 ± 0.02	21.75 ± 10.7	0	5.3 ± 3.7
Total squares crossed	26.25 ± 8.2	17.2 ± 5.7	28.7 ± 9.8	49.9 ± 14.1
Squares in 30 s after				
model presented	—	6.1 ± 2.0	—	19.4 ± 5.6
Tube entry: frequency	1.44 ± 0.7	1.81 ± 0.1	1.44 ± 0.5	3.37 ± 1.0
Tube entry: total				
duration	29.2 ± 14.3	40.2 ± 15.0	6.6 ± 2.9	15.1 ± 5.3
Rearing	7.5 ± 3.6	5.4 ± 2.1	5.8 ± 2.0	14.5 ± 4.5

Note. Results are expressed as means ± SE.

significant effects due to the repeated measure ($F(1, 63) = 10.3, p < .01$). There was no significant difference due to species on this measure, but the interaction term was significant ($F = 4.9, p < .05$). A post-hoc Duncan test showed that this was principally due to before/after differences for *Acomys* ($p < .01$). That is, *Acomys*, more than *Mus*, responded to the presence of the model by increasing the frequency of tube entries.

The duration of tube entry showed significant differences on the repeated measure ($F(1, 63) = 5.7, p < .05$): Both species spend more time in the tube when the model is being presented. Rearing behavior did not show significant effects due to species or to repeated measures, but did show a significant interaction effect ($F = 8.0, p < .01$). A post-hoc Duncan test indicated that this was mainly due to before/after differences for *Acomys* ($p < .01$).

Discussion

It is evident from these results that the two species respond differently to the introduction of the model predator. The initial hypothesis that *Mus* would spend more time in the burrow than would *Acomys* when the model was presented is not directly supported, since both species spent more time in the burrow than they did when the model was not present. However, the model was continuously present during the second half of the testing period, during which time the animals might do a number of things. Inspection of the data to determine the behavioral responses of the animals when the model was first perceived in fact indicated that *Mus* responded initially by using the burrow more than did *Acomys*: Of 16 animals of each species, 7 *Mus* but only 1 *Acomys* entered the burrow immediately after the model was introduced. This is significant with a χ^2 test ($\chi^2 = 4.2, p < .05$). Thus, the *initial* response of *Mus* is indeed to seek cover inside the tube more than does *Acomys*.

Another main species difference seems to be that *Mus* tend to respond more by freezing, whereas *Acomys* tend to exhibit greater activity (as measured by squares crossed and by rearing). *Acomys* also tend to show a somewhat higher frequency of tube entries after the model is presented, although this too may reflect greater activity, rather than increased motivation to enter the tube. This greater activity might be taken to indicate a greater tendency on the part of *Acomys* to respond to the model by fleeing. This is suggested by the data indicating the number of squares crossed in the first 30 s after presentation, which is considerably greater for *Acomys* than for *Mus*. When this measure is compared to the number of squares crossed in the 30 s prior to the model's presentation, then there is little change for *Mus* (mean values of 6.6 before vs 6.1 after). By contrast, there is a dramatic increase for *Acomys* (7.2 before vs 19.4 after—an increase of 271%). Inspection of the raw data for responses when the model was first presented gave further support to this observation: 13 *Acomys* fled, compared to 7 *Mus*.

Fleeing and freezing are alternative avoidance responses; whichever one occurs may depend upon the species, and also upon the circumstances. Fentress (1968), for example, in a study of voles, noted that whether the animal fled or froze depended in part upon the animal's ongoing behavior at the time that the stimulus (an overhead moving object) was presented. However, in the present experiment, the two species were matched for behavior preceding the stimulus as far as possible. Fentress also noted a species difference in the two species used in his study, which he related to underlying differences in an "activity tendency." Thus, the more active *Microtus agrestis* fled more often than did *Clethrionomys britannicus*; similarly, in the present study, the more active *Acomys* fled more often than did the less active *Mus*.

GENERAL DISCUSSION

These experiments have demonstrated a number of differences in exploratory behavior between the two species. In general, the first two experiments indicated that, as predicted, *Mus* is rather more tentative in its exploration of a novel area than is *Acomys*; that is, exploratory behavior of *Mus* is characterized by greater use of the relatively familiar (or possibly safer) area of the start box, and a higher incidence of "ambivalent" behaviors such as peeping out and stretching. The exploratory behavior of *Acomys*, on the other hand, seems to be more decisive, and less ambivalent. The decision to enter the central square, while taking longer for *Acomys*, is not preceded by any overtly ambivalent behavior such as head-poking: The animal goes straight into the central square.

Exploring a novel area carries with it certain costs, particularly in terms of predation risks. Because of this, it might be expected that the exploratory behavior of a small murid rodent would include strategies to minimize these risks, such as the frequent use of available cover from which to base exploratory forays. In addition to this, there may be expected to be species differences, depending upon the habitat to which that species is adapted in the wild. It was noted that *Acomys* for example, behaved differently from *Mus* towards the artificial burrow, treating it more as another kind of novel object, and spending much time investigating the outside of it. *Mus* on the other hand tended to use the burrow more, particularly when startled by a potential predator.

One previous study which compared 12 species of murid rodents in an open field found considerable differences in rearing behavior, such that normally desert-dwelling species showed higher levels of rearing than did grassland dwellers (Wilson et al., 1976). *Acomys* is a desert-dwelling species and, following the suggestion of Wilson et al. that rearing behavior might be a function of living in open areas such as deserts, might be expected to show higher levels of rearing than might *Mus*. However, the present study found no species differences in rearing in

any experiments. Wilson et al. suggested that high levels of rearing in desert animals might be advantageous, in that it might increase the animal's ability to see around it. It is not clear, however, that rearing would be a similar advantage for *Mus*, which normally occupies a range of habitats. Indeed, rearing may even be a disadvantage, since it also makes the animal more conspicuous to predators. In a study of exploratory responses and predation on albino laboratory mice, Glickman and Morrison (1969) found that those that moved about and reared most were more likely to be captured by a resident owl than those that remained in the safety of the home cage. On this basis, the rather high levels of rearing behavior shown by *Mus* would seem to be maladaptive.

One species difference that is worth further note is that the animals seem to adopt different ways of exploring—what we might term different strategies. As noted above, *Mus* are more tentative, seeming to explore outward initially from the start box, and then apparently attending to a variety of stimuli. The impression given by *Acomys* is, however, that it has “somewhere to go.” *Acomys* leave the start box, and then commonly explore the outer square by means of moving around it several times in succession. While doing this, they apparently ignore the central square. After doing this for some time, they then enter the central square and explore that for a time, before returning to the outer square. *Mus* enter the central square sooner, but then make several returns to it during the course of a testing session. These differences clearly need further investigation. However, if they are found to be generally applicable across a range of testing situations, then they may well represent species differences in exploratory strategies. Although, overall, the information gained may be similar, *Acomys* seems to go about it in a more systematic and persistent manner than does the distractible *Mus*.

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